ENTERPRISE CALENDAR INFRASTRUCTURE

Slotbook

A scheduling platform built for zero double bookings under concurrency



PostgreSQL GiST exclusion

Fastify 5 on Node 24

React 19

02 THE PROBLEM

Why check then insert fails

Naive check then insert

A reads: 0 of 1 seat used

B reads: 0 of 1 seat used

A inserts a booking

B inserts a booking

2 bookings for 1 seat

Slotbook

A locks the host row

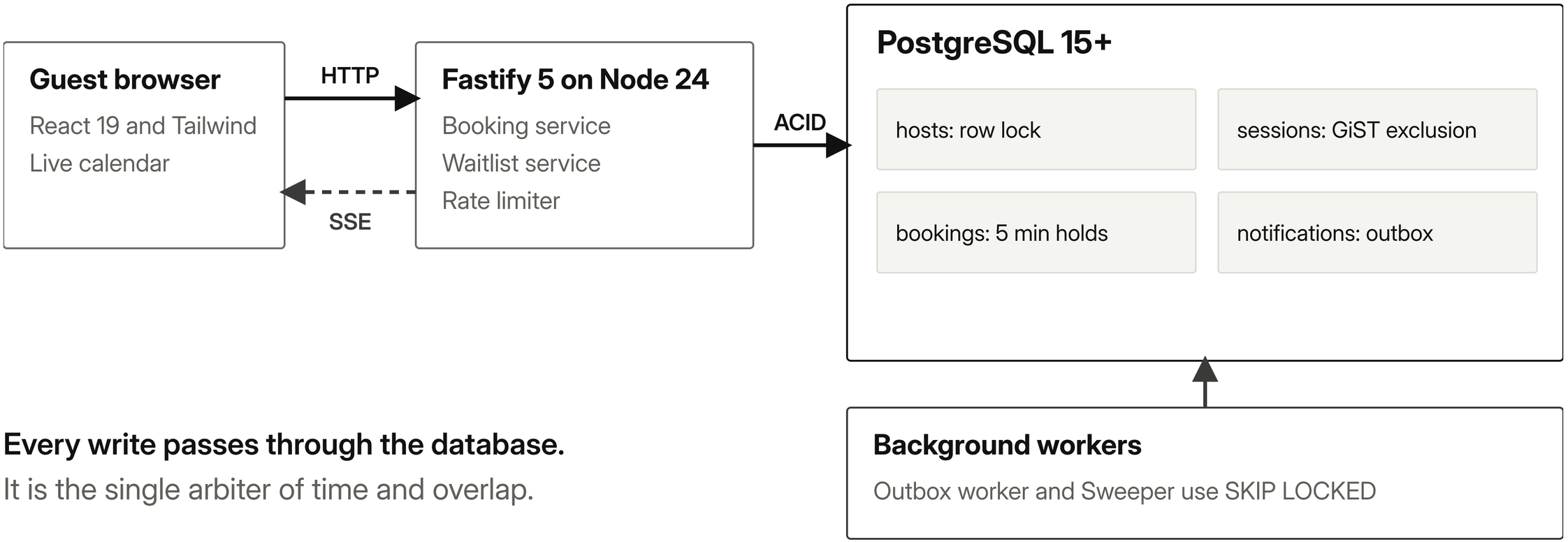
B waits for the lock

A inserts and commits

B gets 23P01 then HTTP 409

1 booking for 1 seat

Three tiers and one arbiter

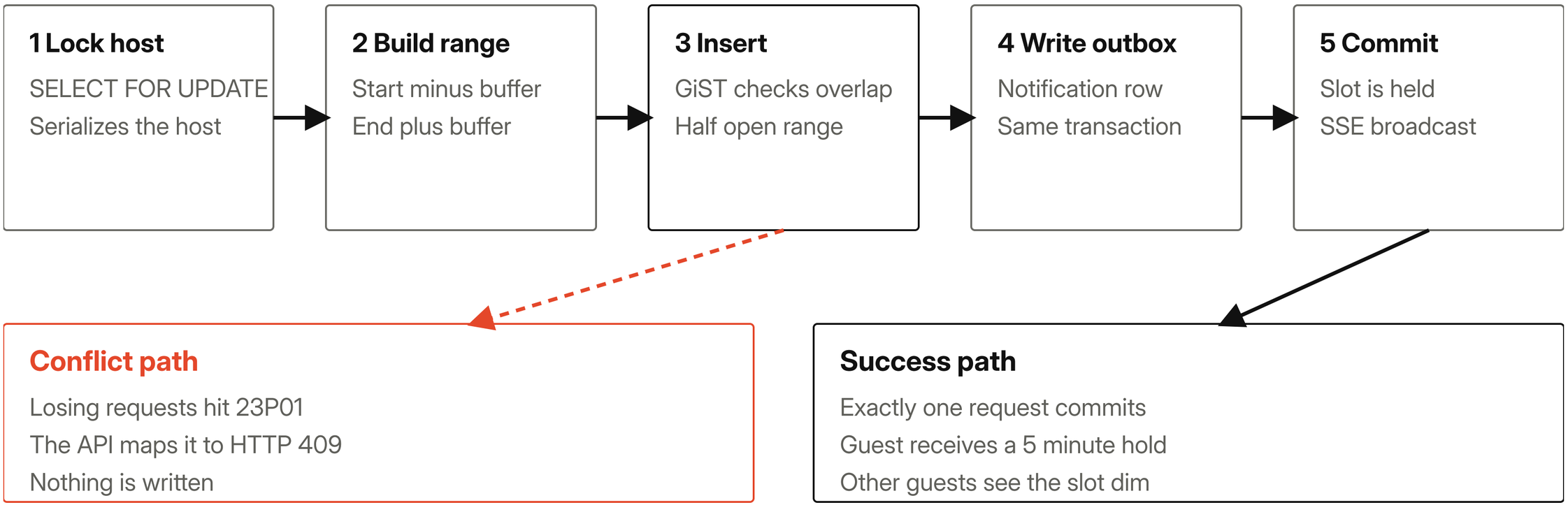


Every write passes through the database.

It is the single arbiter of time and overlap.

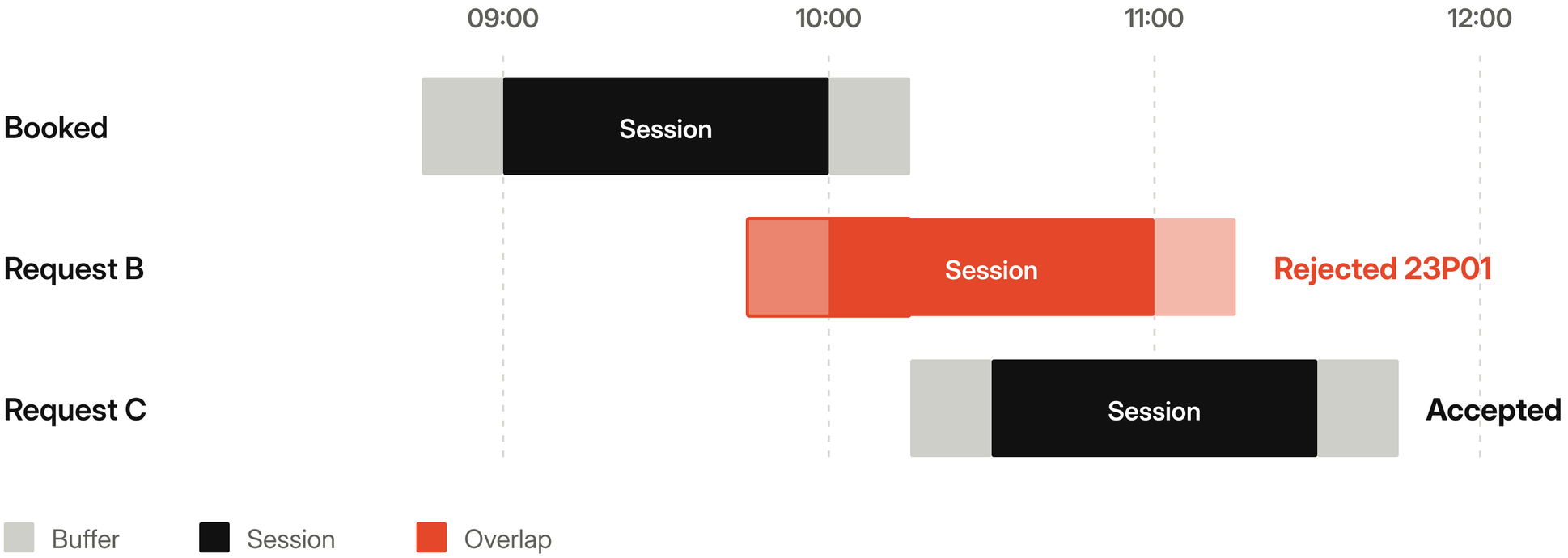
04 BOOKING FLOW

One request through the database



05 OVERLAP PREVENTION

Buffers are part of the reserved range



blocked_range = [start minus buffer before, end plus buffer after)

Main tables and constraints

hosts

id PK, slug UNIQUE

email UNIQUE

timezone IANA

daily_booking_cap

event_types

host_id FK

duration_min, capacity

buffer_before, buffer_after

min_notice, horizon

sessions

blocked_range tstzrange

EXCLUDE USING gist

(host_id =, range &&)

WHERE status active

bookings

session_id FK

status held, confirmed

hold_expires_at

token_hash UNIQUE

notifications

type, payload jsonb

pending, delivered, failed

attempts

next_attempt_at

waitlists

session_id FK

status waiting, offered

hold_id

offered_at

reminders

booking_id FK

type 24h or 1h

scheduled_for

status pending, sent

Supporting tables

weekly_availability

date_overrides

external_busy_times

idempotency_keys

Every range is half open. A violation raises SQLSTATE 23P01 and the API returns HTTP 409.

07 KEY DECISIONS 1 OF 2

Choices that protect correctness

DECISION	ALTERNATIVES CONSIDERED	WHY THIS CHOICE
Row lock plus GiST exclusion	App level count check. SERIALIZABLE with retries. Advisory locks.	The row lock orders each host alone. The constraint is a hard backstop that code cannot bypass.
Database clock as time authority	Application server clock. Client supplied time.	Clock skew cannot extend a hold or skip a notice rule. One clock decides everything.
Buffers inside blocked_range	Separate buffer checks in code. Extra buffer rows.	One range per session lets one constraint protect meetings and prep time together.

08 KEY DECISIONS 2 OF 2

Choices that protect experience

DECISION	ALTERNATIVES CONSIDERED	WHY THIS CHOICE
Transactional outbox	Send the webhook inside the request. Run an external message broker.	Booking and event commit together. No extra infrastructure to operate.
SSE instead of polling	Short polling. WebSockets.	Updates only flow one way. SSE is lighter and removes polling load.
Hold then confirm with release	Book in a single step. Hold with no release.	Guests can inspect a slot safely. Release and the Sweeper free abandoned holds.

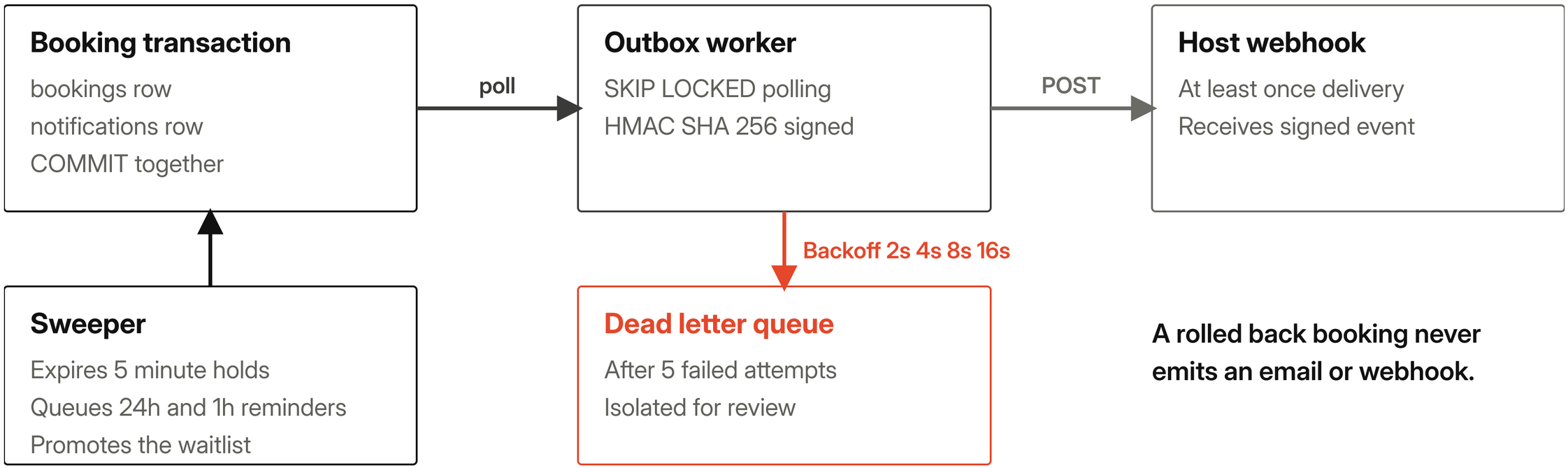
Failure cases: concurrency and time

FAILURE	HOW IT IS HANDLED	TEST THAT PROVES IT
50 guests claim one seat	Host lock serializes. GiST rejects overlap with 23P01. Losers get HTTP 409.	Concurrency suite with 50 simultaneous holds
Buffers collide across meetings	Buffers are part of blocked_range so prep time cannot intersect.	Concurrency suite for buffer collisions. Edge case suite for zero buffer adjacency
Guest clock skew	Hold expiry and notice windows compare against PostgreSQL NOW().	Concurrency suite for hold expiry boundaries
Retry or double click	Idempotency key replays the stored response with no second hold.	Concurrency suite for idempotency replay

Failure cases: lifecycle and delivery

FAILURE	HOW IT IS HANDLED	TEST THAT PROVES IT
Abandoned hold locks a slot	Sweeper expires 5 minute holds. Release endpoint frees them on Back and page unload.	Worker suite for Sweeper cleanup of expired holds and empty sessions
Cancel races with confirm	Both take the host lock so one wins and the other fails cleanly.	Edge case suite for cancellation against confirmation
Webhook endpoint is down	Outbox retries with exponential backoff then moves to the dead letter queue.	Edge case suite for dead letter retry backoff
Worker stops mid delivery	Row stays pending. SKIP LOCKED lets another worker take over.	Worker suite for outbox delivery

Transactional outbox and workers



A rolled back booking never emits an email or webhook.

12 TIME ZONES

Slots are UTC instants

Host wall clock becomes UTC instants. Notice and horizon clamp in UTC. The guest view is a projection.

Example: host in America/New_York with a 01:00 to 04:00 window on 8 March 2026 and 30 minute slots. Guest views from London.

HOST WALL CLOCK	UTC INSTANT	GUEST IN LONDON
01:00 EST	06:00Z	06:00 GMT
01:30 EST	06:30Z	06:30 GMT
02:00 and 02:30	Does not exist. Clocks jump to 03:00	Skipped
03:00 EDT	07:00Z	07:00 GMT
03:30 EDT	07:30Z	07:30 GMT

13 OPTIONAL EXTENSIONS

Each one reuses the core

EXTENSION	HOW IT FITS THE ARCHITECTURE
Waitlist promotion	The waitlist service gives the next guest a 10 minute hold when a booking is cancelled.
24h and 1h reminders	Rows in reminders. The Sweeper dispatches them through the outbox.
Signed webhooks	The outbox worker adds an HMAC SHA 256 signature and logs delivery latency.
External busy sync	Manual blocks and .ics feeds remove overlapping slots in the slot engine.
Calendar export and embed	Guests download .ics or use Google links. An iframe route serves embeds.
Abuse protection and health	Sliding window limit of 150 requests per minute. The health route reports outbox lag.

14 TESTING

47 tests and a load benchmark

47 / 47

Tests pass across 6 files

28 ms

Slot generation average. P95 44 ms for 100 requests

34 ms

Hold placement average. P95 86 ms for 50 holds

0.00%

Unexpected errors under race



14 concurrency | 9 edge cases | 8 extensions | 8 slot engine | 6 api | 2 workers

Where it helped and where it misled

Claude and Gemini sped up DST math, concurrency design, test scaffolding and SSE. They also bloated files. The dashboard went from 1,029 to 267 lines and the booking view from 1,077 to 425. I checked every change against live PostgreSQL.

INCIDENT	WHAT WENT WRONG	MANUAL FIX
SSE stream dropped with ECONNRESET	Fastify closed the raw response. A React effect rebuilt the stream on every slot click.	Added reply.hijack with a 15 second heartbeat. Moved the selected slot into a ref.
Test suites deadlocked with 40P01	Parallel files ran TRUNCATE CASCADE at the same time.	Run files in sequence. Keep the 50 request races inside each suite.
Edge slots vanished for global guests	Notice and horizon were clamped in host local time.	Clamp in UTC instants. Use host wall clock only for daily windows.